

MYELOPROLIFERATIVE NEOPLASMS — One-Pager

Clonal overproduction of MATURE myeloid cells (vs MDS = abnl maturation) · WHO/ELN/NCCN · IM resident reference

OVERVIEW — THE FOUR MPNs

MPN	Cell line ↑	Driver
ET	Platelets	JAK2/CALR/MPL
PV	RBCs (± WBC/plt)	JAK2 V617F (>95%)
PMF	Marrow fibrosis	JAK2/CALR/MPL
CML	Granulocytes	BCR-ABL1 t(9;22)

Any MPN can transform to acute leukemia / marrow failure.

ERYTHROCYTOSIS — USE THE EPO

Type	Problem / examples	EPO
Relative	Hemoconcentration (diuretics, vomiting, diarrhea)	Normal
Secondary	Hypoxia (COPD/OSA/altitude/RAS) or EPO tumor (HCC/RCC/pheo)	High
PV	Autonomous marrow (JAK2)	Low (nl in ~15%)

PV: Hgb >16.5(M)/>16.0(F) or Hct >49/48% + panmyelosis + JAK2.

Splanchnic clot (Budd-Chiari, portal/splenic/mesenteric) → hunt for PV (portal HTN masks labs).

THROMBOCYTOSIS & LEUKOCYTOSIS

Thrombocytosis: reactive [**Fe-def anemia** (often missed), infection/inflam, post-splenectomy] vs primary (MPN). **Magnitude ≠ thrombosis risk**; don't treat reactive. **Leukocytosis:** leukemoid [infection, steroids, stress, tobacco] vs malignant. CML clues: left shift, basophilia/eosinophilia, **low LAP**, WBC >100k (leukostasis rare in CML).

PMF & THE LEUKEMIA TABLE

PMF: marrow fibrosis → extramedullary hematopoiesis; **teardrop cells** + "dry tap" + megakaryocyte hyperplasia.

	Smear / marrow	Unique
AML	Myeloblasts/Auer rods; >20% blasts	DIC (APL)
CML	Eos/baso, WBC 100k; <20% blasts	t(9;22)
ALL	Lymphoblasts; >20%	CNS infiltr (LP)
CLL	Lymphocytosis, smudge cells	Richter, AI cytopenias

TREATMENT PRINCIPLES

MPN	Approach
ET	Prevent thrombosis; risk by age >60/prior clot. Low: observe+ASA. High: +cytoreduction; anticoag if venous, ASA if arterial
PV	Phlebotomy (Hct <45%) + low-dose ASA; +cytoreduction if high risk
PMF	Prognosis-driven; HSCT = only cure; JAK inhibitor for sx
CML	TKI (highly effective); switch for resistance; HSCT rare

2022–25: PV → ropeginterferon (Besremi) NCCN 1st-line cytoreductive. MF JAK inhibitors by counts: ruxolitinib, fedratinib, **pacritinib** (plt<50k), **momelotinib** (anemia). CML → **asciminib** (STAMP inhibitor; ≥2 prior TKIs or T315I).

Adapted from TeachIM.org "Myeloproliferative Neoplasms" (Lakritz, Geiger, McMahon; exec. ed. Zhang; May 2022) · CC BY-NC 4.0 · educational use only, not medical advice · heme-onc therapy evolves — verify & have clinician-reviewed before teaching · optimized June 2026.